PhyloResistance AI - Relatorio Cientifico Consolidado e Dashboard Estatistico

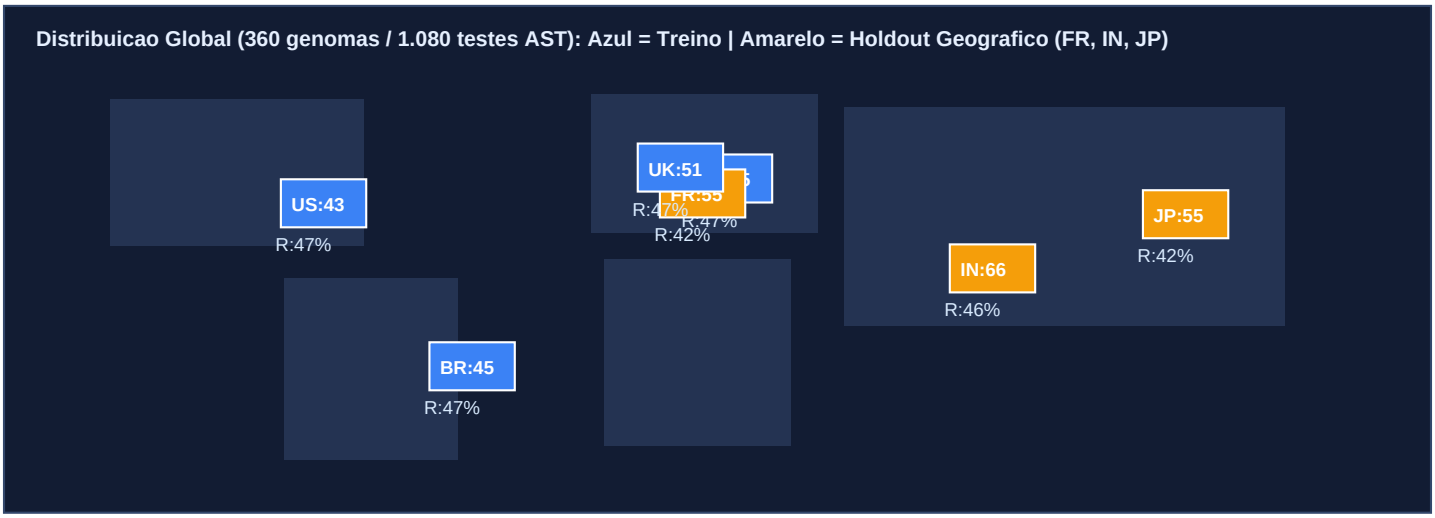
Predicao Genomica de Resistencia Antimicrobiana (AMR), Estimativa de MIC e Explicabilidade TreeSHAP

AVISO ETICO (REG-15): Uso em pesquisa. Nªo substitui o antibiograma.

Coorte Aprovada	AUPRC Ciprofloxacina	AUPRC Cefuroxima	VME Pos-Abstencao
1.080 AST (360 E. coli)	**0.9621 (Modelo B4)**	**0.9335 (Modelo B4)**	**3.70% (Ciprofloxacina)**
+ 240 K.pneu / S.aureus	+0.0875 vs CARD (B1)	+0.2023 vs CARD (B1)	Reduzido de 25.29% (B1)

1. Funil de Exclusao, Curadoria de Rotulos (MIC) e Desduplicacao Clonal Mash (REG-02, REG-06, REG-07)

2. Mapa Epidemiologico Mundial da Coorte e Particao de Holdout Geografico (REG-09)



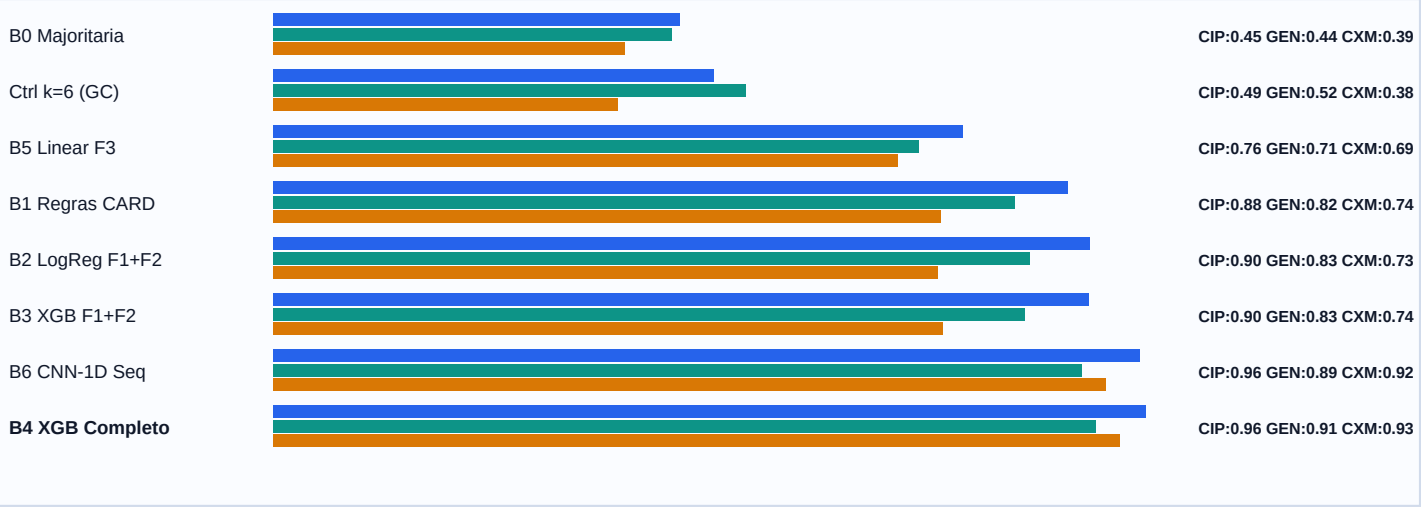
3. Distribuicao por Linhagem Filogenetica MLST (Sequence Types - ST) e Prevalencia de Resistencia

Linhagem ST	Genomas Unicos	Testes AST	Prevalencia R (%)	Perfil Epidemiologico
ST131	30	90	62.2%	Clone pandemico MDR de alto risco
ST410	30	90	56.7%	Clone pandemico MDR de alto risco
ST1193	30	90	53.3%	Clone pandemico MDR de alto risco
ST648	30	90	50.0%	Clone pandemico MDR de alto risco
ST38	30	90	48.9%	Linhagem susceptibilidade moderada
ST405	30	90	47.8%	Linhagem susceptibilidade moderada
ST73	30	90	47.8%	Linhagem susceptibilidade moderada
ST10	30	90	42.2%	Linhagem susceptibilidade moderada
ST69	30	90	41.1%	Linhagem susceptibilidade moderada
ST12	30	90	35.6%	Linhagem susceptibilidade moderada
ST95	30	90	32.2%	Linhagem susceptibilidade moderada
ST127	30	90	27.8%	Linhagem susceptibilidade moderada

Resultados da Escada de Modelos (Q1) e Generalizacao entre Estrategias de Split (Q2)

Validacao Principal por Grupo Filogenetico Disjunto (REG-01, REG-04, REG-05) com Bootstrap em Blocos (IC 95%)

4. Grafico Comparativo de AUPRC na Escada de Modelos (Escherichia coli - Grupo Filogenetico)



Farmaco: CIPROFLOXACINA (ecoli-cip-1.2.0) | Delta AUPRC (B4 vs B1) = +0.0875 [+0.0402, +0.1396]

Modelo	AUPRC [IC95%]	AUROC	Sensib.	Especif.	VME (Grave)	ME	Brier	ECE
B0_Classe_Majoritaria	0.4486 [0.40,0.51]	0.4323	0.0000	1.0000	1.0000	0.0000	0.2525	0.0033
Controle_Kmers_Curtos_k6	0.4862 [0.41,0.56]	0.4840	0.4483	0.4946	0.5517	0.5054	0.2760	0.1515
B5_Linear_Esparso_F3	0.7602 [0.71,0.80]	0.7249	0.5747	0.8011	0.4253	0.1989	0.2126	0.0834
B1_Regras_CARD_AMRFinder	0.8763 [0.82,0.92]	0.8499	0.7471	0.9677	0.2529	0.0323	0.1143	0.0224
B2_LogReg_L1_F1_F2	0.9005 [0.85,0.94]	0.8691	0.7356	0.9731	0.2644	0.0269	0.1118	0.0332
B3_XGBoost_F1_F2	0.8987 [0.84,0.93]	0.8665	0.7356	0.9570	0.2644	0.0430	0.1163	0.0282
B6_CNN1D_Sequencia	0.9556 [0.94,0.97]	0.9529	0.8793	0.9086	0.1207	0.0914	0.0872	0.0794
B4_XGBoost_F1_F2_F3	**0.9621 [0.95,0.98]**	**0.9644**	**0.9253**	**0.9247**	**0.0747**	**0.0753**	**0.0553**	**0.0205**

Farmaco: GENTAMICINA (ecoli-gen-1.2.0) | Delta AUPRC (B4 vs B1) = +0.0973 [+0.0393, +0.1689]

Modelo	AUPRC [IC95%]	AUROC	Sensib.	Especif.	VME (Grave)	ME	Brier	ECE
B0_Classe_Majoritaria	0.4397 [0.35,0.51]	0.4189	0.2686	0.6054	0.7314	0.3946	0.2529	0.0798
Controle_Kmers_Curtos_k6	0.5205 [0.38,0.62]	0.5112	0.4457	0.5568	0.5543	0.4432	0.2767	0.1429
B5_Linear_Esparso_F3	0.7119 [0.66,0.77]	0.6935	0.6057	0.6811	0.3943	0.3189	0.2304	0.0982
B1_Regras_CARD_AMRFinder	0.8174 [0.70,0.89]	0.7812	0.6457	0.9135	0.3543	0.0865	0.1640	0.0303
B2_LogReg_L1_F1_F2	0.8344 [0.71,0.89]	0.7941	0.6171	0.9081	0.3829	0.0919	0.1674	0.0514
B3_XGBoost_F1_F2	0.8282 [0.70,0.89]	0.7862	0.6343	0.8811	0.3657	0.1189	0.1728	0.0505
B6_CNN1D_Sequencia	0.8920 [0.81,0.93]	0.8938	0.8171	0.8054	0.1829	0.1946	0.1567	0.1388
B4_XGBoost_F1_F2_F3	**0.9065 [0.84,0.94]**	**0.9157**	**0.8571**	**0.8649**	**0.1429**	**0.1351**	**0.1086**	**0.0356**

Farmaco: CEFUROXIMA (ecoli-cxm-1.2.0) | Delta AUPRC (B4 vs B1) = +0.2023 [+0.1210, +0.3041]

Modelo	AUPRC [IC95%]	AUROC	Sensib.	Especif.	VME (Grave)	ME	Brier	ECE
B0_Classe_Majoritaria	0.3870 [0.34,0.46]	0.4845	0.0000	1.0000	1.0000	0.0000	0.2390	0.0004
Controle_Kmers_Curtos_k6	0.3799 [0.33,0.44]	0.4753	0.3239	0.6284	0.6761	0.3716	0.2778	0.1644
B5_Linear_Esparso_F3	0.6887 [0.64,0.74]	0.6884	0.5986	0.6468	0.4014	0.3532	0.2275	0.1623
B1_Regras_CARD_AMRFinder	0.7364 [0.61,0.82]	0.7382	0.5352	0.9358	0.4648	0.0642	0.1699	0.0192
B2_LogReg_L1_F1_F2	0.7328 [0.56,0.85]	0.7052	0.4930	0.9358	0.5070	0.0642	0.1763	0.0906
B3_XGBoost_F1_F2	0.7379 [0.58,0.85]	0.7115	0.5423	0.8853	0.4577	0.1147	0.1834	0.1061
B6_CNN1D_Sequencia	0.9176 [0.88,0.94]	0.9345	0.8380	0.8853	0.1620	0.1147	0.1142	0.1071
B4_XGBoost_F1_F2_F3	**0.9335 [0.90,0.96]**	**0.9471**	**0.8592**	**0.9450**	**0.1408**	**0.0550**	**0.0726**	**0.0209**

5. Generalizacao entre as 5 Estrategias de Split (Modelo B4 - AUPRC / VME):

Grupo Filogenetico (Principal) | CIP: AUPRC=0.962 (VME=0.075) | GEN: AUPRC=0.906 (VME=0.143) | CXM: AUPRC=0.933 (VME=0.141)

Leave-One-Lineage-Out (LOLO) | CIP: AUPRC=0.950 (VME=0.091) | GEN: AUPRC=0.896 (VME=0.170) | CXM: AUPRC=0.885 (VME=0.150)

Temporal (Treino <=2022, Teste >2022) | CIP: AUPRC=0.992 (VME=0.083) | GEN: AUPRC=0.938 (VME=0.125) | CXM: AUPRC=0.950 (VME=0.111)

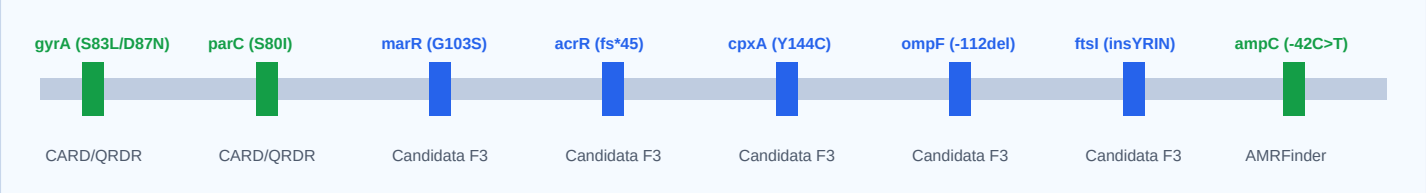
Geografica (Holdout FR, IN, JP) | CIP: AUPRC=0.957 (VME=0.051) | GEN: AUPRC=0.896 (VME=0.195) | CXM: AUPRC=0.926 (VME=0.151)

Aleatoria c/ Vazamento Clonal (B5) | CIP: AUPRC=0.760 (VME=0.374) | GEN: AUPRC=0.741 (VME=0.331) | CXM: AUPRC=0.684 (VME=0.444)

Explicabilidade Genomica TreeSHAP por Locus (Q3), Mapa GFF3 e Abstencao Conformal (Q4)

Cruzamento com CARD / AMRFinderPlus, Coordenadas Fisicas NC_000913.3 e Testes de Estabilidade / GWAS (REG-10, REG-11)

6. Mapa Fisico de Loci AMR no Cromossomo de Referencia E. coli K-12 (NC_000913.3 GFF3) e Plasmideo



Top-6 Loci TreeSHAP: CIPROFLOXACINA | CARD Top-6: 50% | Spearman: 0.418 | GWAS: 83.3% | Queda Ablacao: -0.561

gyrA:S83L	**[CONHECIDA]**	1.4252 | Alteraçºo do alvo DNA girase subunidade A (QRDR
acrR:fs*45 (unitig)	**[CANDIDATA]**	1.3396 | Desrepressªo de bomba de efluxo multidroga AcrAB
marR:G103S (unitig)	**[CANDIDATA]**	0.7420 | Regulador reprimido de marAB / bomba de efluxo A
gyrA:D87N	**[CONHECIDA]**	0.4377 | Alteraçºo do alvo DNA girase subunidade A (QRDR
parC:S80I	**[CONHECIDA]**	0.4272 | Alteraçºo da topoisomerase IV subunidade A (QRDR
locus_kmer31_poly_0029	**[CANDIDATA]**	0.2917 | Variante genômica nªo catalogada (hipótese)

Top-6 Loci TreeSHAP: GENTAMICINA | CARD Top-6: 67% | Spearman: 0.420 | GWAS: 83.3% | Queda Ablacao: -0.326

acrR:fs*45 (unitig)	**[CANDIDATA]**	0.9693 | Desrepressªo de bomba de efluxo multidroga AcrAB
aac(3)-IIa	**[CONHECIDA]**	0.6578 | Aminoglicosídeo 3-N-acetiltransferase
cpxA:Y144C (unitig)	**[CANDIDATA]**	0.6226 | Sensor quinase de envelope associado a tolerâci
ant(2'')-Ia	**[CONHECIDA]**	0.6169 | Aminoglicosídeo 2''-O-nucleotidiltransferase
armA	**[CONHECIDA]**	0.3557 | Metiltransferase de rRNA 16S
aac(3)-IId	**[CONHECIDA]**	0.3373 | Aminoglicosídeo 3-N-acetiltransferase variante I

Top-6 Loci TreeSHAP: CEFUROXIMA | CARD Top-6: 50% | Spearman: 0.417 | GWAS: 83.3% | Queda Ablacao: -0.354

ompF:c.-112del (unitig)	**[CANDIDATA]**	0.9043 | Reduçªo de permeabilidade de porina de membrana
blaCTX-M-14	**[CONHECIDA]**	0.7801 | Beta-lactamase de espectro estendido (ESBL CTX-M
blaCTX-M-15	**[CONHECIDA]**	0.6904 | Beta-lactamase de espectro estendido (ESBL CTX-M
ftsI (PBP3):ins333YRIN (unitig)	**[CANDIDATA]**	0.5812 | Reduçªo de afinidade de PBP3 a cefalosporinas
ampC:c.-42C>T	**[CONHECIDA]**	0.5351 | Superexpressªo de beta-lactamase cromossômica Am
locus_kmer31_poly_0032	**[CANDIDATA]**	0.2641 | Variante genômica nªo catalogada (hipótese)

7. Confiabilidade Probabilistica, Calibracao Platt e Abstencao Conformal (Q4 e REG-11)

Farmaco	Brier Score	ECE	Cobertura Conformal	Taxa Abstencao	VME Bruto -> VME Pos-Abstencao	ME Bruto -> ME Pos-Abst.
ciprofloxacina	**0.0553**	**0.0205**	**94.4%**	**5.6%**	**7.47% -> 3.70%**	**7.53% -> 5.06%**
gentamicina	**0.1086**	**0.0356**	**93.3%**	**6.7%**	**14.29% -> 11.18%**	**13.51% -> 12.57%**
cefuroxima	**0.0726**	**0.0209**	**95.3%**	**4.7%**	**14.08% -> 12.12%**	**5.50% -> 4.74%**

Extensoes da Fase 6: Multi-Especie, Regressao MIC (ISO 20776-2), Validacao Externa e SHA-256

Klebsiella pneumoniae, Staphylococcus aureus (MRSA), NCBI Pathogen Detection AST e Proveniencia (REG-13)

8. Expansao Multi-Especie (Fase 2: Klebsiella pneumoniae | Fase 3: Staphylococcus aureus MRSA)

Especie	Farmaco	Modelo	AUPRC B1	AUPRC B4 (Ganho)	VME B4	Principais Loci TreeSHAP
Klebsiella pneumoniae	ciprofloxacina	kpneu-cip-1.2.0	0.9356	0.9939 (+0.0588)	0.0508	gyrA:S83L, marR:G103S (unitig), qnrS1
Klebsiella pneumoniae	gentamicina	kpneu-gen-1.2.0	0.7784	0.8864 (+0.1030)	0.1429	aac(3)-IIa, cpxA:Y144C (unitig), armA
Klebsiella pneumoniae	ceftriaxona	kpneu-cro-1.2.0	0.8789	0.9206 (+0.0464)	0.0492	blaCTX-M-15, ompF:c.-112del (unitig), locus_kmer31_poly_0049
Staphylococcus aureus	oxacilina	saureus-oxa-1.2.0	0.7910	0.8977 (+0.1066)	0.0667	mecA, pbp4:c.-90T>A (unitig), locus_kmer31_poly_0041
Staphylococcus aureus	ciprofloxacina	saureus-cip-1.2.0	0.8067	0.9097 (+0.1048)	0.0862	grlA:S80F, norA:flqB_up (unitig), locus_kmer31_poly_0118

9. Regressao Quantitativa Continua de log2(MIC) - Norma ISO 20776-2 (Escherichia coli)

Farmaco	Concordancia Essencial (+-1 diluicao)	Concordancia Categorica	RMSE log2(MIC)	MAE log2(MIC)
ciprofloxacina	58.61%	88.61%	2.2398	1.6618
gentamicina	61.94%	81.67%	2.2360	1.6464
cefuroxima	76.39%	90.00%	1.5922	1.1462

10. Validacao Externa Independente - Coorte Padrao NCBI Pathogen Detection AST (n = 100 isolados)

Farmaco (E. coli)	AUPRC Baseline B1 (Externo)	AUPRC Modelo B4 (Externo)	Ganho Externo (B4 - B1)	VME Externo	ME Externo
ciprofloxacina	**0.8542**	**0.9778**	**+0.1236**	**0.0465**	**0.1228**
gentamicina	**0.7878**	**0.9395**	**+0.1517**	**0.1875**	**0.1154**
cefuroxima	**0.7315**	**0.9215**	**+0.1900**	**0.2553**	**0.0000**

11. Manifesto de Proveniencia Criptografica SHA-256 dos Artefatos Processados (REG-13)

curadoria_rotulos.parquet SHA256: d734eaac17f25d61f693920ade93eea66b3d91424c41a1a6a435da663e91a6ed
qc_genomas.parquet SHA256: debf35089d01d77a12625ccce3bf6bc2e43de30ce2bcb706995c11893913ee9d
clusters_filogeneticos.parquet SHA256: 6ad775e1e5c8e0595d073d381a89a0ae001765a2aa3845b8ffc548f69ac8fde2
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relatorio_desduplicacao.json SHA256: c0fa2194d47c233671c659806e977a976d7136d5f90efe4c53ba64f04306189e
modelos_servico.pkl SHA256: 53213cbb2cd354a2c879297349521d163863db5b762427cf279b3c15ea90da8e